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Ex.No 9: Massive MIMO Capacity vs SNR and Gain Using MATLAB

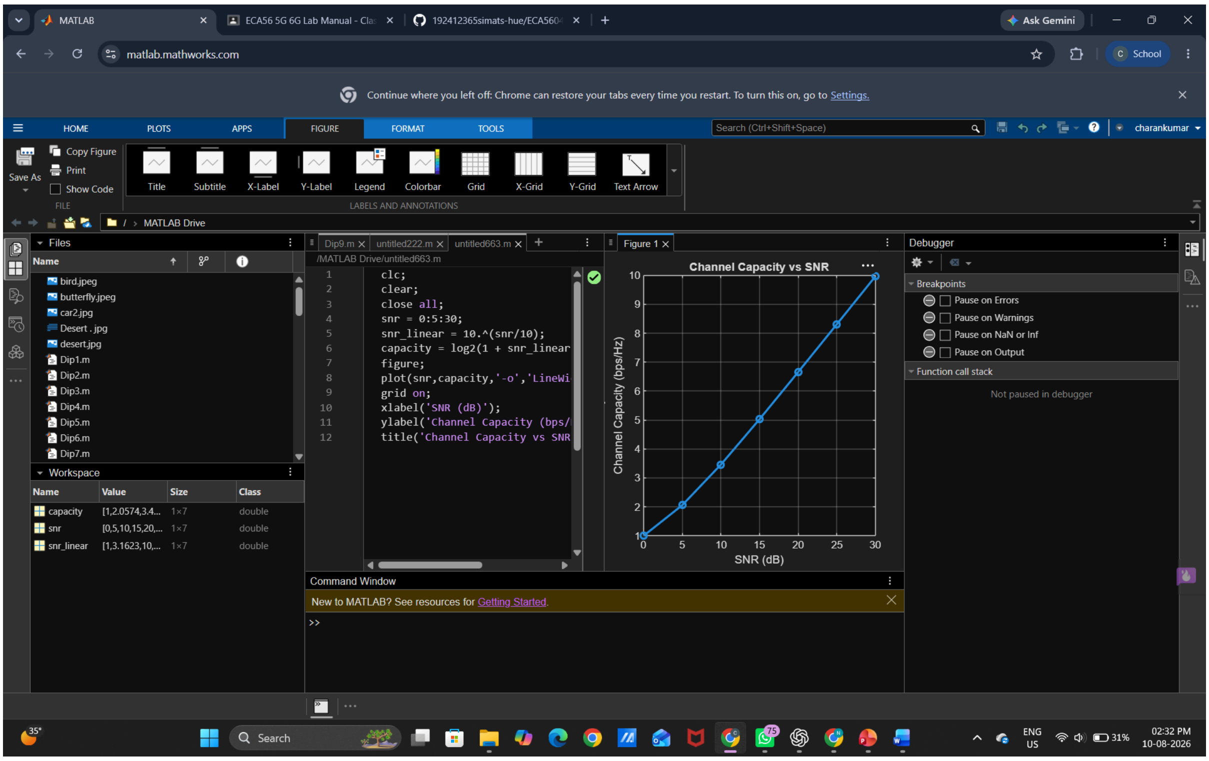
Objective 1:

Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio (SNR) values using MATLAB. Observe how channel capacity increases with improving signal quality based on the Shannon capacity principle..

Program (Matlab Code)

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

Output:



Objective 2:

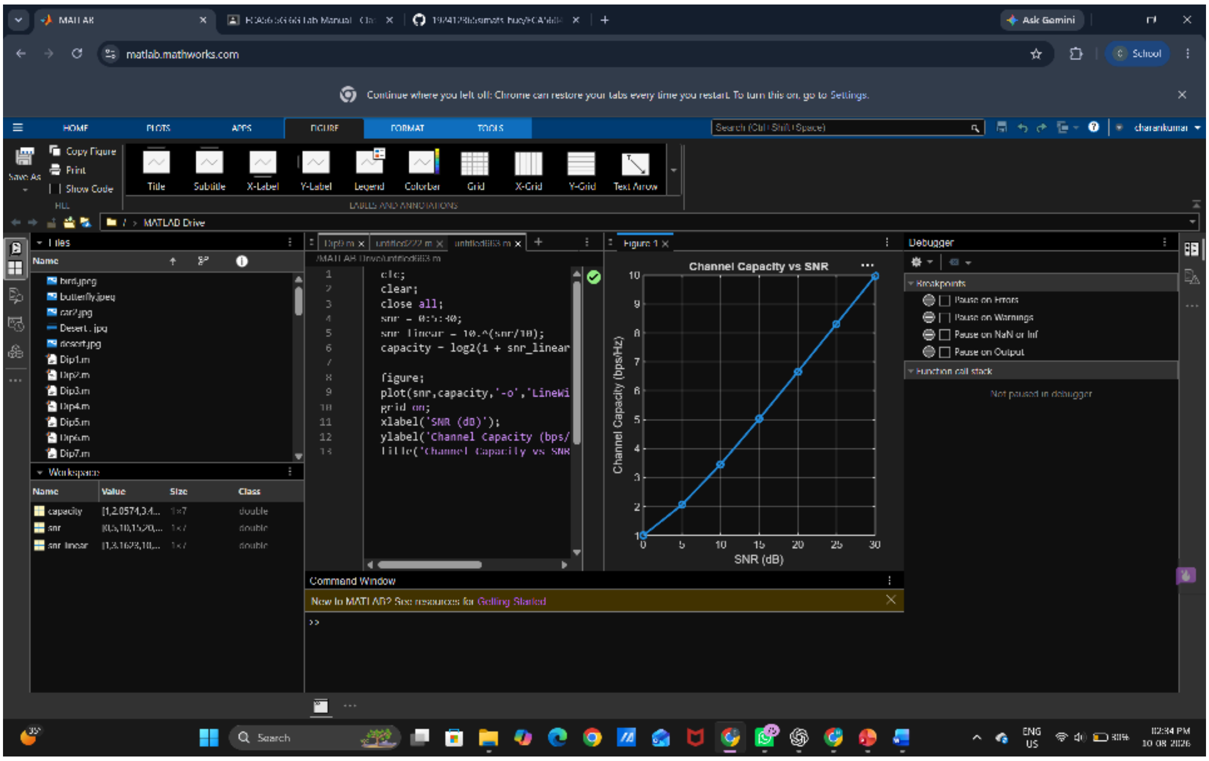
Compare the signal-to-noise ratio for different communication scenarios using MATLAB Analyze the effect of SNR on the quality and reliability of wireless communication in Massive MIMO systems...

Matlab Code

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);

figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

Output



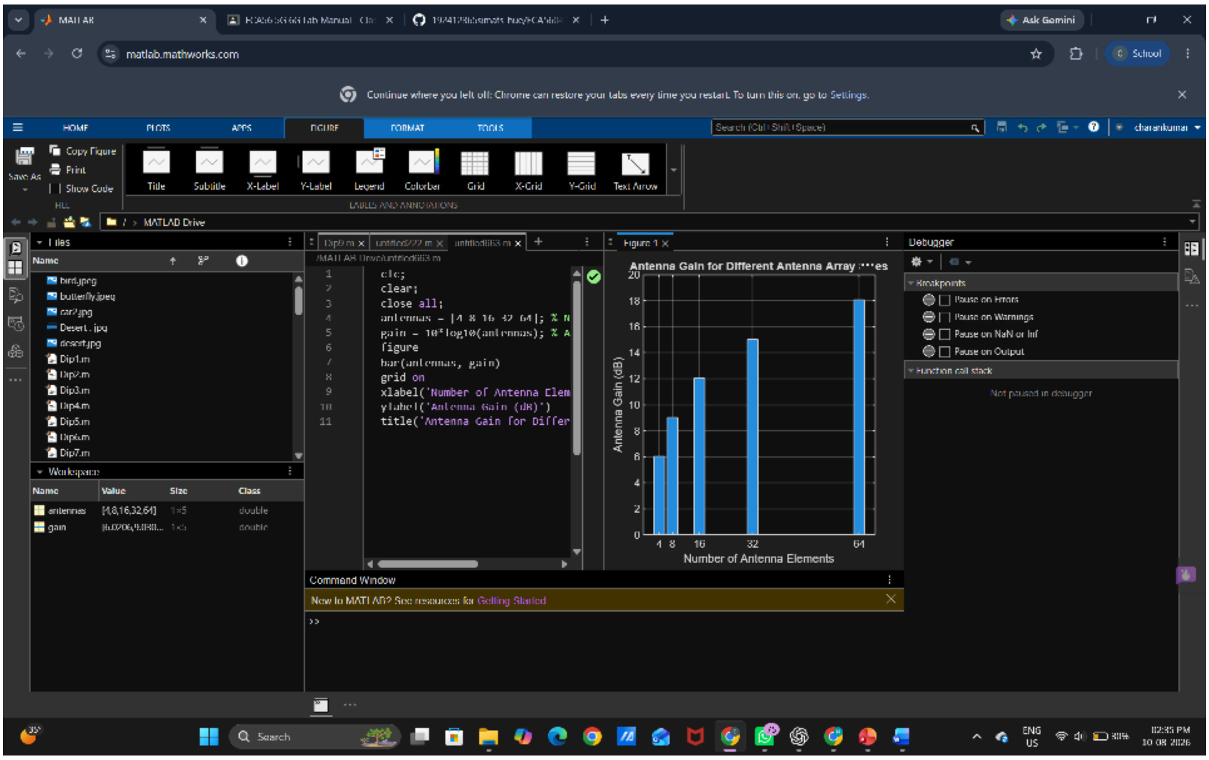
Objective 3:

Analyze the antenna gain achieved by varying the number of antenna elements in a Massive MIMO system using MATLAB. Observe how increasing the antenna array size improves directional gain and enhances communication performance..

Matlab Code

```
clc;
clear;
close all;
antennas = [4 8 16 32 64]; % Number of Antenna Elements
gain = 10*log10(antennas); % Antenna Gain (dB)
figure
bar(antennas, gain)
grid on
xlabel('Number of Antenna Elements')
ylabel('Antenna Gain (dB)')
title('Antenna Gain for Different Antenna Array Sizes')
```

Output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car7.jpg
desert.jpg
Dip1.m
Dip2.m
Dip3.m
Dip4.m
Dip5.m
Dip6.m
Dip7.m
Workspace
Name
Value
Size
Class
antennas
[4,8,16,32,64]
1x5
double
gain
1x5
double
clc;
clear;
close all;
figure
bar(antennas, gain)
grid on
Figure 1
Antenna Gain for Different Antenna Array
Antenna Gain (dB)
Number of Antenna Elements
Debugger
Breakpoints
Pause on Errors
Pause on Warnings
Pause on NaN or Inf
Pause on Output
Function call stack
Not paused in debugger
Command Window
New to MATLAB? See resources for Getting Started
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